Stationary weighted Schiffer: perturbation on the ball

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1 Where we are

The note `direction2-parabolic-to-classical.tex` proved that the lateral parabolic Schiffer condition for a weighted pair (Ω, V) reduces to classical Schiffer for the unweighted Laplacian on the base. That closes the parabolic angle on Direction 2 (and makes precise what the de Dios Pont barrel limit can and cannot do). The residual question is the *stationary* weighted Schiffer problem:

Definition 1 (Stationary weighted Schiffer pair)**.** *A smooth convex pair (Ω, V) with eigenpair (ϕ, λ) is a* stationary weighted Schiffer pair *if*

$$L_{\Omega,V}\phi := -\Delta\phi + \nabla V \cdot \nabla\phi = \lambda\phi \text{ in } \Omega, \quad \partial_\nu\phi = 0 \text{ on } \partial\Omega, \quad \phi|_{\partial\Omega} \equiv c, \quad \phi \not\equiv 0. \tag{1}$$

The ball with any radial convex V produces a countable family of such pairs (one per radial Langevin eigenvalue). The natural extension of the Schiffer conjecture is:

Conjecture 2 (Log-concave Schiffer)**.** *If (Ω, V) is a smooth convex stationary weighted Schiffer pair, then Ω is a ball and V is radial (with respect to the centre of Ω).*

This generalizes the classical Schiffer conjecture (recovered for $V \equiv 0$) and is the right target for Direction 2 with the log-concave compactification.

Below we attack the conjecture perturbatively at the unweighted endpoint $V = 0$ on the ball, and identify the precise codimension of the Schiffer-preserving directions of V. The output is:

- Schiffer-preserving first-order deformations V_1 on the ball form an infinite-codimensional but still infinite-dimensional linear subspace; the codimension is enumerated by an explicit family of linear functionals $\mathcal{L}_{\ell,m}$ indexed by nonzero spherical harmonics (ℓ, m), each of which is a one-dimensional Green-function moment of $V_{1,\ell,m}$ against the radial profile ϕ_0'.
- The radial directions $V_1 = V_1(r)$ are all in this kernel, matching the known continuum of radial weighted Schiffer pairs.
- The infinitesimal log-concave Schiffer rigidity question on the ball reduces to: *does every first-order Schiffer-preserving deformation extend to all orders, modulo $O(n)$ symmetries?* A positive answer would say the perturbation is forced into the radial direction.

2 Setup

Throughout this note $\Omega = B = B_R \subset \mathbb{R}^n$, $n \geq 2$. Fix a radial Neumann eigenfunction

$$\phi_0(r) \quad \text{with} \quad -\Delta\phi_0 = \lambda_0\phi_0 \text{ in } B, \quad \phi_0'(R) = 0,$$

i.e., ϕ_0 is a Bessel-type radial eigenfunction at a radial Neumann eigenvalue λ_0. By radiality $\phi_0(R) = c_0$ is automatically a constant boundary value, so $(B, 0, \phi_0, \lambda_0)$ is a (trivial) weighted Schiffer pair.

Perturb with $V_\varepsilon = \varepsilon V_1$, $V_1 \in C^\infty(\overline{B})$, and expand $\phi_\varepsilon = \phi_0 + \varepsilon\phi_1 + O(\varepsilon^2)$, $\lambda_\varepsilon = \lambda_0 + \varepsilon\lambda_1 + O(\varepsilon^2)$.

3 First-order eigenvalue equation

Lemma 3. (ϕ_1, λ_1) *satisfies*

$$(-\Delta - \lambda_0)\phi_1 = \lambda_1\phi_0 - \nabla V_1 \cdot \nabla\phi_0 \text{ in } B, \qquad \partial_\nu\phi_1 = 0 \text{ on } \partial B, \tag{2}$$

and the Fredholm condition gives

$$\lambda_1 = \frac{1}{\|\phi_0\|^2_{L^2(B)}} \int_B (\nabla V_1 \cdot \nabla\phi_0)\,\phi_0\,dx = \frac{1}{2\|\phi_0\|^2_{L^2(B)}} \int_B \nabla V_1 \cdot \nabla(\phi_0^2)\,dx. \tag{3}$$

Proof. Substitute the expansion into $L_{B,\varepsilon V_1}\phi_\varepsilon = \lambda_\varepsilon\phi_\varepsilon$ and collect the $O(\varepsilon)$ term. The Neumann BC at first order is $\partial_\nu\phi_1 = 0$ since both ϕ_0 and ϕ_ε have $\partial_\nu = 0$ on ∂B. Self-adjointness of $-\Delta - \lambda_0$ with Neumann data on its kernel span$\{\phi_0\}$ yields (3). $\square$

4 Decomposition into spherical harmonics

Let $\{Y_{\ell,m}\}_{\ell\geq 0,m}$ be the standard orthonormal basis of $L^2(S^{n-1})$ with $-\Delta_{S^{n-1}}Y_{\ell,m} = \ell(\ell+n-2)Y_{\ell,m}$. For $\ell = 0$ take $Y_{0,1} \equiv |S^{n-1}|^{-1/2}$. Decompose

$$V_1(x) = \sum_{\ell,m} V_{1,\ell,m}(r)\,Y_{\ell,m}(\hat{x}), \qquad \phi_1(x) = \sum_{\ell,m} \Phi_{\ell,m}(r)\,Y_{\ell,m}(\hat{x}).$$

Since ϕ_0 is radial, $\nabla V_1 \cdot \nabla\phi_0 = \phi_0'(r)\partial_r V_1 = \phi_0'(r)\sum_{\ell,m} V'_{1,\ell,m}(r)Y_{\ell,m}(\hat{x})$. Write the radial Laplacian in the ℓ-th sector as

$$\Delta_r^{(\ell)}u(r) := u''(r) + \tfrac{n-1}{r}u'(r) - \tfrac{\ell(\ell+n-2)}{r^2}u(r).$$

Equation (2) separates:

$$(-\Delta_r^{(\ell)} - \lambda_0)\Phi_{\ell,m}(r) = \delta_{\ell,0}\,\lambda_1\,\phi_0(r)\,\delta_{m,1}|S^{n-1}|^{1/2} - \phi_0'(r)\,V'_{1,\ell,m}(r), \tag{4}$$

with $\Phi'_{\ell,m}(R) = 0$ and regularity at $r = 0$.

5 The Schiffer-preserving constraint at first order

The first-order weighted Schiffer condition is $\phi_\varepsilon|_{\partial B} \equiv c_\varepsilon$, which gives $\phi_1(R\hat{x}) = c_1$ for every unit vector $\hat{x} \in S^{n-1}$. In spherical-harmonic modes:

$$\Phi_{0,1}(R) = c_1\,|S^{n-1}|^{1/2}, \qquad \Phi_{\ell,m}(R) = 0 \quad \text{for all } (\ell, m) \text{ with } \ell \geq 1. \tag{5}$$

The $\ell = 0$ equation just adjusts the boundary constant by εc_1, which is allowed (the Schiffer condition only demands constancy in the angular variable, not a prescribed value); so $\ell = 0$ imposes no constraint on V_1. The genuine constraint is on the $\ell \geq 1$ modes.

Proposition 4 (First-order rigidity functional)**.** *For each $\ell \geq 1$ and each m, the constraint $\Phi_{\ell,m}(R) = 0$ is equivalent to the vanishing of the linear functional*

$$\mathcal{L}_{\ell,m}(V_1) := \int_0^R G_\ell(R, r')\, \phi_0'(r')\, V_{1,\ell,m}'(r')\, dr', \tag{6}$$

where $G_\ell(R, r')$ is the Green's function on $[0, R]$ for the operator $-\Delta_r^{(\ell)} - \lambda_0$ with regular boundary behaviour at $r = 0$ and Neumann condition $u'(R) = 0$, evaluated at $r = R$.

Proof. For $\ell \geq 1$ the right-hand side of (4) is just $-\phi_0'(r)V_{1,\ell,m}'(r)$. Solve by convolution against the Green's function and read off the value at $r = R$. $\square$

Remark 5 (Existence and non-degeneracy of G_ℓ)**.** *For generic radial eigenvalues λ_0, λ_0 is not a Neumann eigenvalue of $-\Delta_r^{(\ell)}$ for $\ell \geq 1$, so G_ℓ exists. When λ_0 accidentally lies in the ℓ-sector Neumann spectrum a finite-codimensional Fredholm adjustment is required, but generically does not occur for the radial Bessel eigenvalues used here.*

Corollary 6 (The first-order kernel)**.** *The space of $V_1 \in C^\infty(\overline{B})$ such that $(B, \varepsilon V_1)$ admits a first-order Schiffer-preserving extension of (ϕ_0, λ_0) is the linear subspace*

$$\mathcal{K} := \bigcap_{\ell \geq 1,\, m} \ker \mathcal{L}_{\ell,m} \subset C^\infty(\overline{B}). \tag{7}$$

6 Radial perturbations and the size of $\mathcal{K}$

Corollary 7 (Radial directions are admissible)**.** *Every radial $V_1 = V_1(r)$ lies in $\mathcal{K}$.*

Proof. For radial V_1, $V_{1,\ell,m} \equiv 0$ for all (ℓ, m) with $\ell \geq 1$, so $\mathcal{L}_{\ell,m}(V_1) = 0$ trivially. $\square$

Corollary 8 (Beyond radial)**.** *$\mathcal{K}$ strictly contains the radial subspace. In particular, for every (ℓ, m) with $\ell \geq 1$, the kernel $\ker \mathcal{L}_{\ell,m}$ inside the angular sector $V_{1,\ell,m}$ is infinite-codimension-one in $C^\infty([0, R])$; the joint intersection over (ℓ, m) remains infinite-dimensional.*

Proof. $\mathcal{L}_{\ell,m}$ is a single nonzero linear functional on the infinite-dimensional space of radial profiles $V_{1,\ell,m}$, so its kernel is of codimension 1 in that sector. Different (ℓ, m) sectors are independent, so the joint kernel intersects each sector in a codimension-1 subspace. $\square$

This says the first-order Schiffer-preserving perturbations are an infinite-dimensional family that strictly contains the radial directions. If Conjecture 2 is true, all but the radial directions must be ruled out by higher-order obstructions or by removing $O(n)$-trivial directions.

7 $O(n)$-trivial directions

The $O(n)$-action on B produces a finite-dimensional family of "trivial" perturbations that do not change the pair up to symmetry. Each angular harmonic $Y_{\ell,m}$ for $\ell = 1$ is generated by an infinitesimal translation of the centre of B (a rigid motion): specifically, $V_1(x) = x_j$ corresponds to translating the centre of the sphere. Including these in the analysis: the $\ell = 1$ sector of $\mathcal{K}$ is exactly the span of $\{x_j \cdot g(r) : g \in \ker \mathcal{L}_{1,j}\}$, and modulo translations of the centre this sector is reduced by n dimensions.

Higher $\ell \geq 2$ harmonics are not generated by rigid motions, so their kernel $\ker \mathcal{L}_{\ell,m}$ is genuinely infinite-dimensional beyond symmetries.

This is the key observation: *infinitesimal log-concave Schiffer rigidity on the ball is equivalent to: for every* $\ell \geq 2$ *and every* $V_{1,\ell,m} \in \ker \mathcal{L}_{\ell,m} \setminus \{0\}$*, the formal Lyapunov–Schmidt series for* $(\phi_\varepsilon, \lambda_\varepsilon, V_\varepsilon)$ *fails to converge to a genuine weighted Schiffer pair at some finite order* ≥ 2*.*

8 The second-order test

The natural follow-up computation is the second-order Schiffer condition. Write $\phi_\varepsilon = \phi_0 + \varepsilon\phi_1 + \varepsilon^2\phi_2 + \cdots$. The $O(\varepsilon^2)$ equation is

$$(-\Delta - \lambda_0)\phi_2 = \lambda_2\phi_0 + \lambda_1\phi_1 - \nabla V_1 \cdot \nabla\phi_1. \tag{8}$$

At order 2, the Schiffer condition $\phi_2(R\hat{x}) = c_2$ gives, for each (ℓ, m) with $\ell \geq 1$,

$$\int_0^R G_\ell(R, r')\big[\lambda_1\Phi_{\ell,m}(r') - (\nabla V_1 \cdot \nabla\phi_1)_{\ell,m}(r')\big]\,dr' = 0, \tag{9}$$

which involves the angular-product structure $(V_1 \cdot \phi_1)_{\ell,m}$. This is a quadratic functional in V_1. The key analytic question is:

> *Is there a non-radial* $V_1 \in \mathcal{K}$ *such that* (9) *holds for all* (ℓ, m) *with* $\ell \geq 1$*, and similarly all higher-order constraints?*

A positive answer produces a genuine non-radial branch of weighted Schiffer pairs on the ball, falsifying Conjecture 2 at the perturbative level. A negative answer (which the conjecture predicts) requires showing that the quadratic constraint (9) together with the linear $\mathcal{L}_{\ell,m} = 0$ system has only the radial solutions.

9 Next steps

1. **Compute** $G_\ell(R, r')$ **explicitly** for the disc ($n = 2$) and ball ($n = 3$) using Bessel and trigonometric representations, and check whether $\mathcal{L}_{\ell,m}$ has sign-definite Green-function kernel against ϕ_0'. If so, $\mathcal{L}_{\ell,m}(V_{1,\ell,m})$ controls $V_{1,\ell,m}'$ moments in a useful way.

2. **Check** $\ell = 1$ **explicitly.** The $\ell = 1$ sector is where rigid-motion symmetries live; the corresponding kernel should be exactly the translation directions modulo symmetry. A direct check confirms whether the first-order analysis "sees" the rigid-motion freedom correctly.

3. **Quadratic test for** $\ell = 2$**.** The smallest non-trivial angular harmonic at which the conjecture would be tested is $\ell = 2$. Carry out (9) for $V_1 = V_{1,2,m}(r)Y_{2,m}$ in $\ker \mathcal{L}_{2,m}$ and determine whether any non-zero such V_1 is consistent.

4. **Connect to known classical results.** The second-order test is, when V_1 is taken instead as an infinitesimal boundary perturbation of the disc (not a potential perturbation), the standard Lyapunov–Schmidt analysis for classical Schiffer near the disc (Direction 3 of the original notes). The relationship between these two perturbations should illuminate which features of the Schiffer obstruction come from geometry of $\partial\Omega$ versus from the potential V.